
Embedded Systems International

Project Statement of Work (PD3)

Team Identifier: G6

Team Name (optional): Nyx

Team Member Names: Olivia Blais, Kaylee Johnson, Rishita Kollu, Lavanya Vangapandu

Submit your document as a PDF file in Canvas under the corresponding project assignment.

One of your section's lab TAs must approve of this Statement of Work by adding a comment in the Canvas assignment. The team is responsible for requesting approval promptly after submission.

Refer to the Project Requirements document before completing this Statement of Work (SOW). A statement of work is a focused concise proposal and agreement that describes work to be done. Teams should complete and submit this SOW form, which represents several parts of a statement of work, including a plan for what you are doing and how. The SOW defines the scope of your project and the approach you are taking to deliver on the goals.

Problem Statement

First, has your team reached consensus on the autonomous vehicle (AV) application you will use as the context or story for your project? All projects, regardless of application, will need to meet the same basic requirements and will be recognized for innovative features.

To define your problem, think about one or more users and their needs. Write at least one Point of View (POV) statement for your application. Follow Steps 2 and 3 in the following guide (also in Canvas as a PDF document):

[Define and Frame Your Design Challenge](#) (links to IDF webpage)

URL: <https://www.interaction-design.org/literature/article/define-and-frame-your-design-challenge-by-creating-your-point-of-view-and-ask-how-might-we>

See the Lab Project page in Canvas for the PDF file for the webpage.

Next, think about and write a few sentences that give a high-level summary of the broad mission goals for your AV application. For example:

The purpose of the AV is to provide

The AV will be capable of doing

The mission goals and user needs establish the purpose of the project and why you are working on it. Now you should translate these into a more detailed problem statement that provides a specific, concise, clear and thorough description of the context for the problem, an explanation of user needs that will be addressed, and an outline of your proposed technical approach to solving the problem.

Problem Statement :

The AV will aid night guards in monitoring and securing buildings at night.

POV statement:

Venues, Institutional, and commercial clients need enhanced and consistent security at night because security guards are typically understaffed or prone to being exhausted, and this increases the chances of a threat with no reliable monitoring.

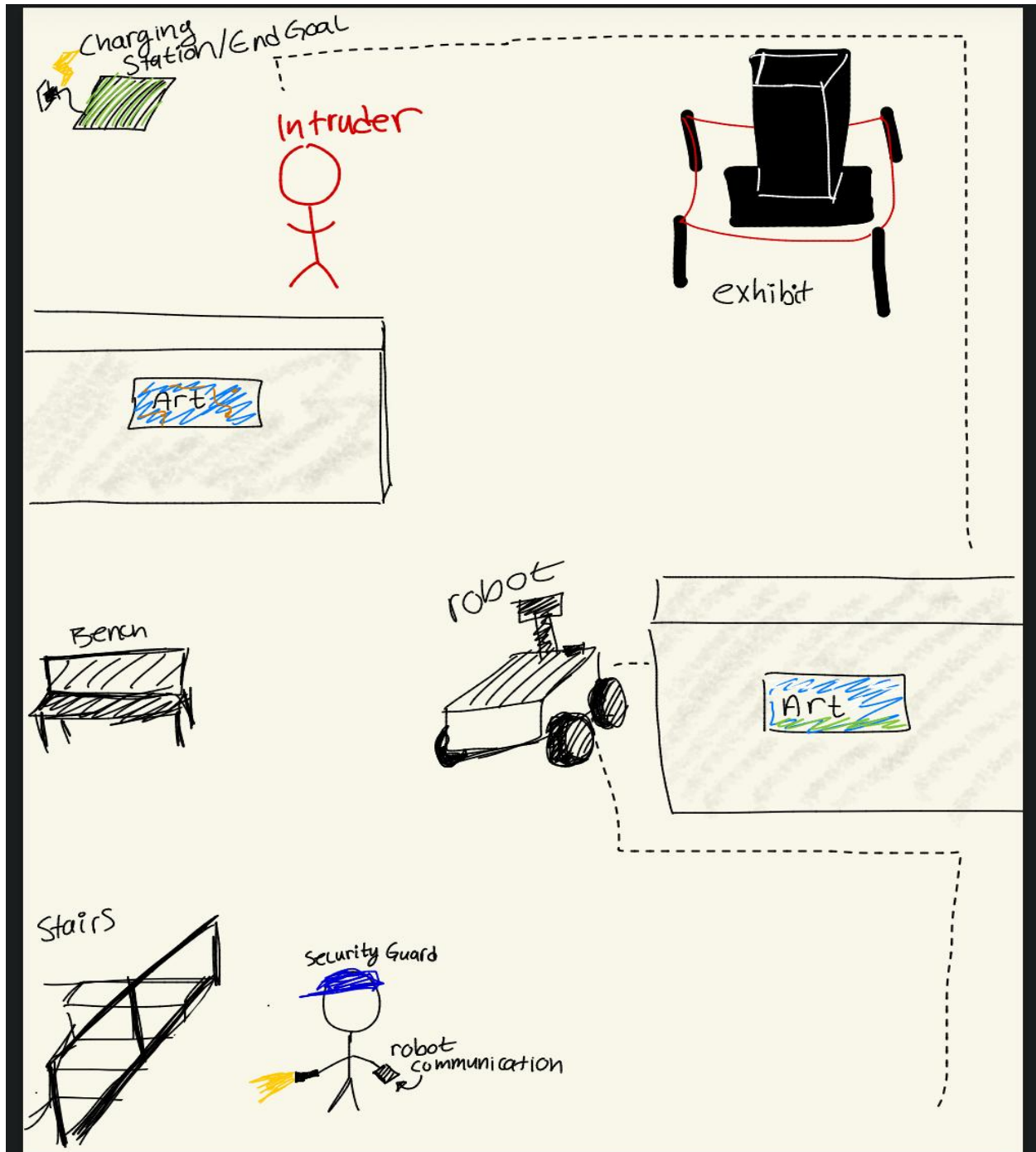
High-level mission goals:

The AV will monitor a designated area, which could be a room in a museum or office, or a marked area with indicative tape laid out on the floor to signal the end of its search area. The AV will look for intruders in the building at night to aid the night guards, making it possible for them to monitor the whole building from the security room's computer.

Short description of the problem:

Night is the time most criminals break into buildings, whether to vandalize property or rob the business. Thus, night guards are hired to monitor the building to ensure these criminals don't get in. However, despite the night guards knowing they will be working at night, they have lives that also must be lived during the day, so they may need to be awake during the day as well as at night. This may make the guards lethargic and slow to respond. The AV will make it so that businesses may not need as many night guards to secure a building. The AV will monitor a designated area and send information back to the security room's computer, allowing fewer night guards to monitor larger areas at once. So smaller businesses with fewer night guards can feel protected and safe, and larger businesses will not need as many guards on the premises each night. This would also let some night guards have more days off for their personal lives.

In addition to writing a paragraph about the problem, you are to draw a **problem sketch: a one-page sketch illustrating your solution with a user context (big picture view)**. This should show the scope of your work in relation to one or more user needs. Refer to the sample project sketches.



Design Approach

Next, consider your AV application in relation to the project requirements and the five categories by which it will be evaluated.

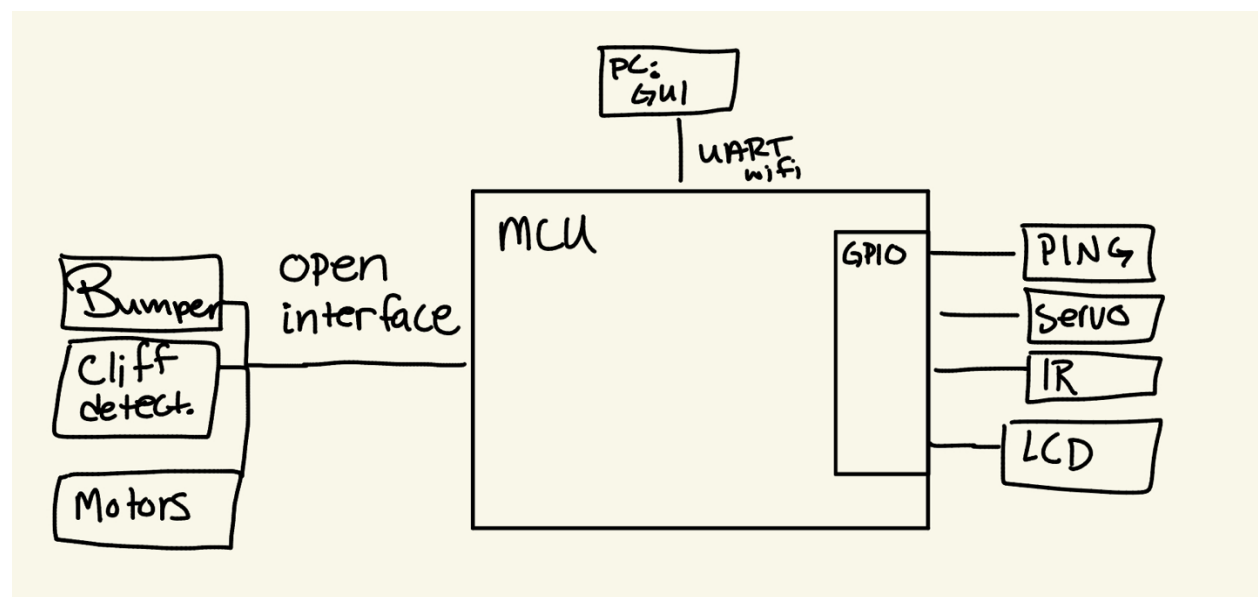
- 1) Functionality in relation to the AV application mission goals and user needs
- 2) Mapping of functional requirements to platform components and capabilities
- 3) Elements of the test field
- 4) Serious incident penalties
- 5) Feature bonuses

In this section, you will identify and describe how you will design your application for each of these categories.

To complete the tables below, your team may want to use ideation tools, such as Lotus Blossom. Refer to the Lab Project page in Canvas and see the project ideas guidelines and Lotus Blossom worksheet as needed.

In addition to completing the tables below, you are to draw a **technical system sketch: a one-page sketch depicting a high-level technical system diagram of your proposed solution, such as a block diagram or dataflow diagram**. This should show both hardware and software modules.

The tables below are your initial proposal, and you may update these before your demonstration.



Functionality

Describe each of the basic functionalities required for the project in terms of your AV application. **The functionality must be specific to the problem and user(s).** Several functional statements are given in the example functional description for the Mars rover application in the Project Requirements document.

Basic Functionality	Mapping to AV Application
Cybot Communication	notifies night guards when an intruder is detected within the area
Cybot Movement	Patrols the area by sweeping up and down its designated area. It may deviate from the patrol path when an intruder is detected and may be interrupted for manual control. The robot will then resume the patrol using orientation tracking.
Object Detection	utilizes external sensors to navigate through a building/room and look for intruders at night
Object Avoidance	During the bot's sweep, it will need to be proactive to avoid museum exhibits and avoid getting stuck, no matter the floor layout.
Boundary Adherence	will move within set boundaries, either a room or an area that would be marked off with indicative tape.
Arrival at Destination	The guard will interrupt the robot once security is no longer needed, and will find its docking station. Will start "charging" once it enters its docking/charging station in the morning
User Interface	Night guards would be able to monitor the building using the sensors mounted/embedded into the AV and see where it is in the room/area where its guarding via a map created by with its sensors.
Base Station Control	The guards are allowed to remotely control the robot by interrupting its auto-patrol and directing it to inspect the area of interest.

Other Application Specific Functionality (may be novel features for bonus points)

When the robot detects a potential intruder, it interrupts its patrol path to investigate the area. After the threat is cleared (either autonomously or by security), the robot will use the gyroscope orientation to realign itself with its previous path and resume its patrol from the correct position and direction.

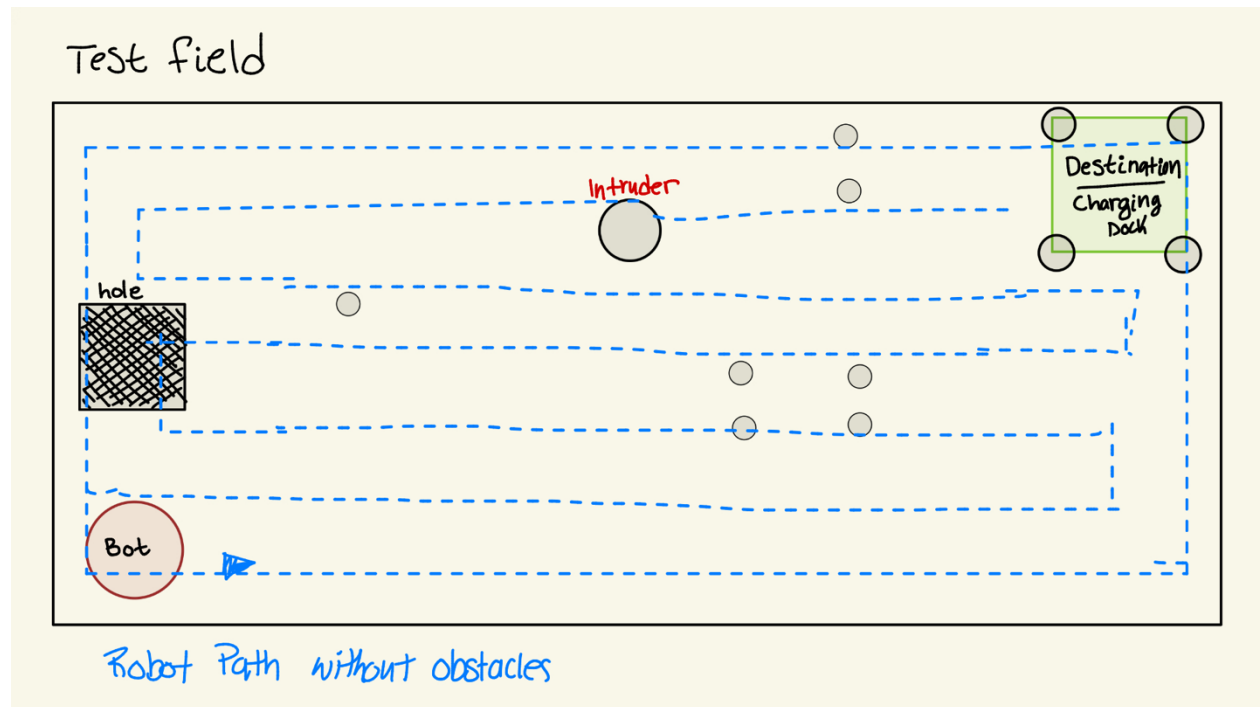
Mapping to Platform

Briefly describe how each of the basic platform components required for the project will be used in your AV application.

Basic Platform Components	Usage in AV Application
Open Interface and iRobot Sensors	OI for movement (setting wheels, turning), iRobot sensors to detect falling into holes (wheel drop) and bumping into short objects (bump sensors).
Interrupts	Receiving intruder alerts
ADC	Detect the intruders and obstacles
Input Capture	Used for ping to measure object or intruder distance
PWM	Control the servo and its scan speed
UART/WiFi	Transmitting the intruder alerts or abnormal activity
Other Platform Components or Modes (may be novel features for bonus points)	
Use the gyroscope for orientation tracking after manual interruptions.	

Elements of the Test Field

Briefly describe a test field in the context of the real application (e.g., Martian terrain, city streets, etc.). Then state what each of the basic objects and other elements required for the test field represent in terms of the AV application. Draw and attach a **sketch of a possible simple test field for the lab**.



Test Field Description

The test field in the context of our application represents an indoor environment, such as a large room, a section of a building, or a hallway, where the robot will perform its security patrol. It is designed to simulate the conditions the robot will face during its surveillance. The intruder, whom the bot is designed to detect, will be represented by the tall, large object on the test field. The hole will represent uneven terrain in the building, such as a step down or large air vents, in which the robot may fall into or get stuck on due to its wheels. The short objects and thin pillars will represent the exhibits, tables, or chairs. The boundaries of the test field are the robot's designated patrol area, where it must adhere.

Basic Objects and Other Elements	Mapping to AV Application Test Field
Tall objects (wide or composite)	Intruders
Short objects	Ledge/stair/step/table base/ storage boxes
Holes	stairs/vent/hole in floor/ uneven flooring
Pillars (thin tall object)	Table leg/stanchions/ barriers
Out of bounds	Outside of the designated room/marked area
Destination zone	Docking/Charging station
Other Application Specific Elements (may be novel features for bonus points or incidents to avoid) Moving intruder (person walking slowly instead of static object) Low lighting conditions (to simulate nighttime security) Random obstacle placement (to simulate real building changes)	

Serious Incidents to Avoid and/or Novel Features (Optional)

You may have identified novel features in the tables above. Enter them in the table below and propose possible bonus points if demonstrated successfully. In addition, describe any additional serious incidents that might happen in your test field for your AV application.

Novel Features Using the Gyroscope to get back on it's patrol path Identifying multiple intruders	Bonus Points +5 +5
--	---------------------------------

Serious Incidents Misidentifying objects as intruders (false alarm)	Deductions -3
---	-------------------------

Sketches

Attach the following sketches to your submission. These were noted above in red.

- **Problem sketch**
- **Technical system sketch**
- **Test field sketch**

Visual communication is helpful for sharing information. There are many ways to represent your information. The problem sketch could be anything from a cartoon (informal) to a UML use case diagram (more formal). The technical system sketch is similar to what has been used/shown in class and lab. The test field sketch probably needs no explanation.